7. Design Brief — TCC Event Clearance App

Nov 8, 2026 · @rogiie

The interface must work in bright sunlight, in a moving line, on low-end phones. Every activity card is a template the SBO designs per event.

Design principles

1. **Readable in sunlight.** High contrast, large type, dark text on light backgrounds, touch targets of at least 48 px.
2. **One job per screen on event day.** The scanner has no menus competing for attention.
3. **Color means one thing.** Green is accepted, amber needs the SBO to decide, red is invalid. Every color also has an icon and a word, never color alone.
4. **Always show the state.** An offline badge, "X scans waiting to sync", and the venue and slot banner stay visible.
5. **Light on low-end phones.** Compressed images, system fonts, no heavy animation.
6. **Plain words.** English first, Filipino labels later, and every error says what to do next.
7. **Private by default.** A student screen never shows another student's data.

Screen inventory

Role	Screens
Everyone	Install gate, Open-in-browser notice, Google sign-in, Consent
Student	Registration form, Live photo, Waiting for approval, Event list, Event pack download, QR screen, My card, Retake photo, Device reset request, Help
SBO officer	Pending registrations, Pending resets, Signature pad, Event list, Event builder (days and slots, venues map, signers, card design), AI artwork picker, Card preview, Scanner prep, Scanner, Scan result, Manual check-in, Check card, Sync status and backup export
Head officer	Officer management, Publish event, Scanner keys

Role	Screens
Web dashboard (officers)	Attendance table with filters, Edit record, Export and print, Disputes, Audit log

Scanner screen specification

The scanner is the screen used most, in a moving line, with one hand.

1. **Status strip (top).** Online or offline, "ready for offline" in green, and "X scans waiting to sync".
2. **Venue and slot banner.** Large text, for example "Scanning at: Location 2 · Day 1 Morning In". Tap to change. Forgetting to change it is the most likely mistake, so it must be impossible to miss.
3. **Camera view.** About half the screen, with a frame to aim at.
4. **Result card.** The student's photo takes at least 40 percent of the screen width, with the name in large type and department, year and section below. Flags show as amber chips (location, not eligible, already scanned, clock off).
5. **Reject button.** Always reachable with the thumb. A green result is saved automatically after about 1.5 seconds unless the SBO taps Reject; an amber result waits for an explicit Accept or Reject. Reject shows reason chips: photo is not a face, does not match.
6. **Bottom bar.** Manual check-in, Check card and Sync.

States: waiting for a QR, accepted (green, short vibration), flagged (amber), invalid or expired (red, with the reason in words).

Card template system

Every card is data, not a fixed design. SBOs change the design for each event, so nothing about layout, wording, number of days or number of slots is hardcoded. The card editor edits the event's `layout_json`.

Element	What the SBO can change
Text	Content, font, size, color, alignment, position

Element	What the SBO can change
Student field	Which field (name, year and section, school ID, department), label, position
Logo or image	Upload, position, size, opacity (a watermark is an image with low opacity)
AI artwork	Choose one option, fit, opacity
Day header	Label such as "DAY 1", fill color
Signature slot	Label (Time In, Time Out or custom), signer, printed name and position under the line
QR	Position and size of the card's QR
Line or shape	Stroke, fill

Rules

- Page size is selectable (for example short bond, A4 or A5), and the card prints from the dashboard at that size.
- The school logo and header text start as defaults the SBO may change.
- Each event keeps its own template version, so a new design never alters an old card.
- A new event can start from a copy of any earlier template, or from a preset.
- AI draws only background artwork; every word on the card is real text.

Sample template: CIT Intramurals 2026

This is the first template, read from the photo you sent. It is a starting point; you will change it.

- **Header, top center:** the school logo, then "Tagoloan Community College", "College of Information Technology" and "Intramurals 2026".
- **Title:** "ATTENDANCE".
- **Student fields:** NAME on the left and YEAR & SECTION on the right, each with a line.
- **Day strips:** five dark gray strips labeled DAY 1 to DAY 5.
- **Slots:** four signature lines per day, labeled Time In, Time Out, Time In, Time Out (morning and afternoon).

- **Signer under each line:** printed name and position. The photo shows different officers on different days (SBO Auditor on the earlier days, SBO Executive Vice Governor on the later ones), so signers are set per slot.
- **Watermark:** the CIT mascot artwork, faint, behind the whole card.

Template data for this card: 5 days, 4 slots per day, 2 student fields, 1 watermark, 20 signature slots with signers set per slot.

Please send the clearance or activity card formats from other events so I can check this system handles them without changes.

Visual style and accessibility

- **Palette:** light background, near-black text, one accent color the school can change, plus the three status colors. Default to light mode because screens are used in sunlight.
- **Type:** system fonts for speed; scanner names at 24 px or larger, body text at 16 px or larger.
- **Contrast:** meet WCAG AA (4.5 to 1 for body text).
- **Touch targets:** at least 48 px, with space between Accept and Reject so a thumb cannot hit the wrong one.
- **Feedback:** short vibration and an optional sound on accept; status never depends on color alone.
- **Motion:** minimal. The only animation that matters is the live card's moving element in the app, so a frozen screenshot stands out.
- **Install guide:** pictures for each step on iPhone and Android, since many students will do this alone.
- **Language:** English now, with all labels in one file so Filipino can be added.